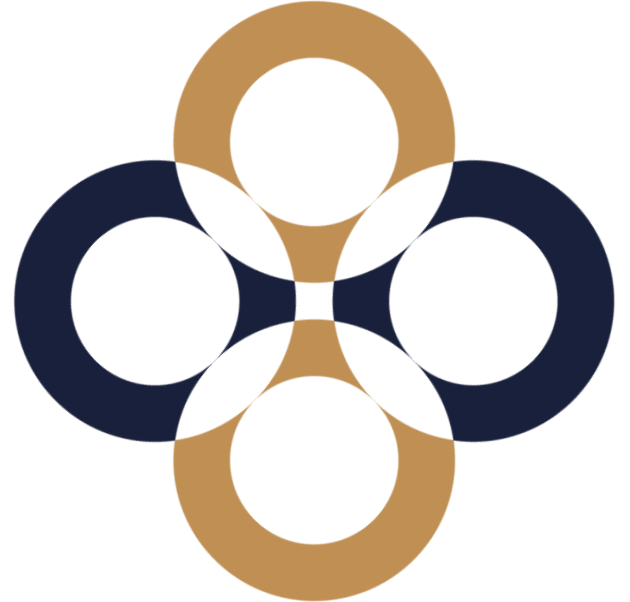
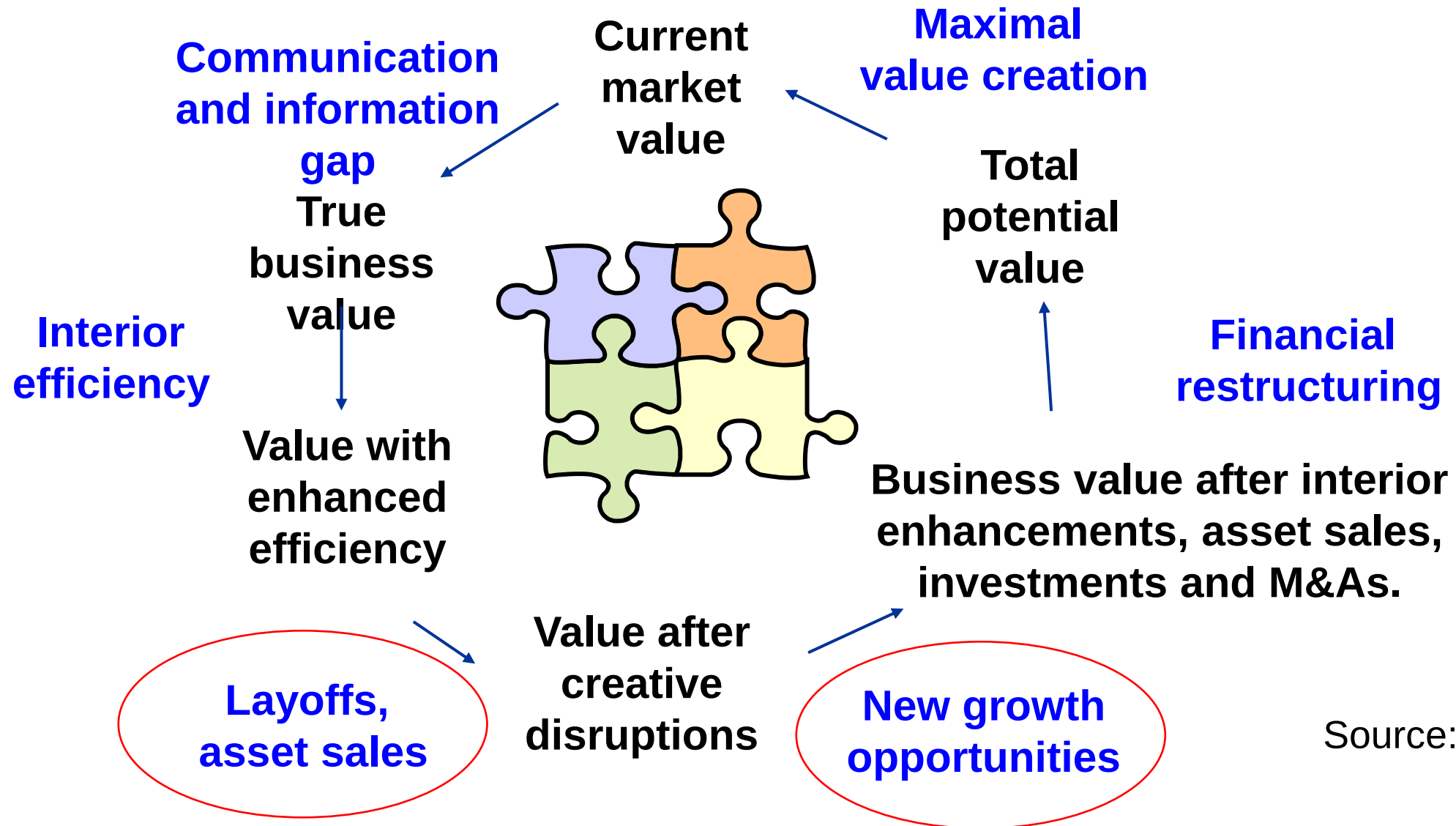


Mergers and Acquisitions

Péter Juhász, PhD, CFA



Value creation areas

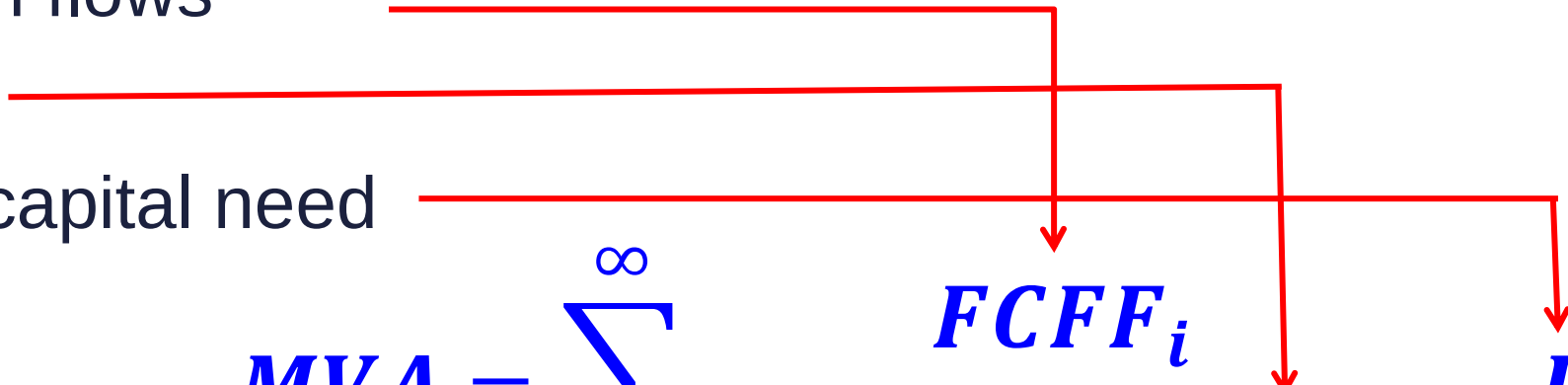


Ways of creating value

Increase cash flows

Reduce risks

Cut back on capital need


$$MVA = \sum_{i=1}^{\infty} \frac{FCFF_i}{\prod_{j=1}^i (1 + WACC_j)} - IC_0$$

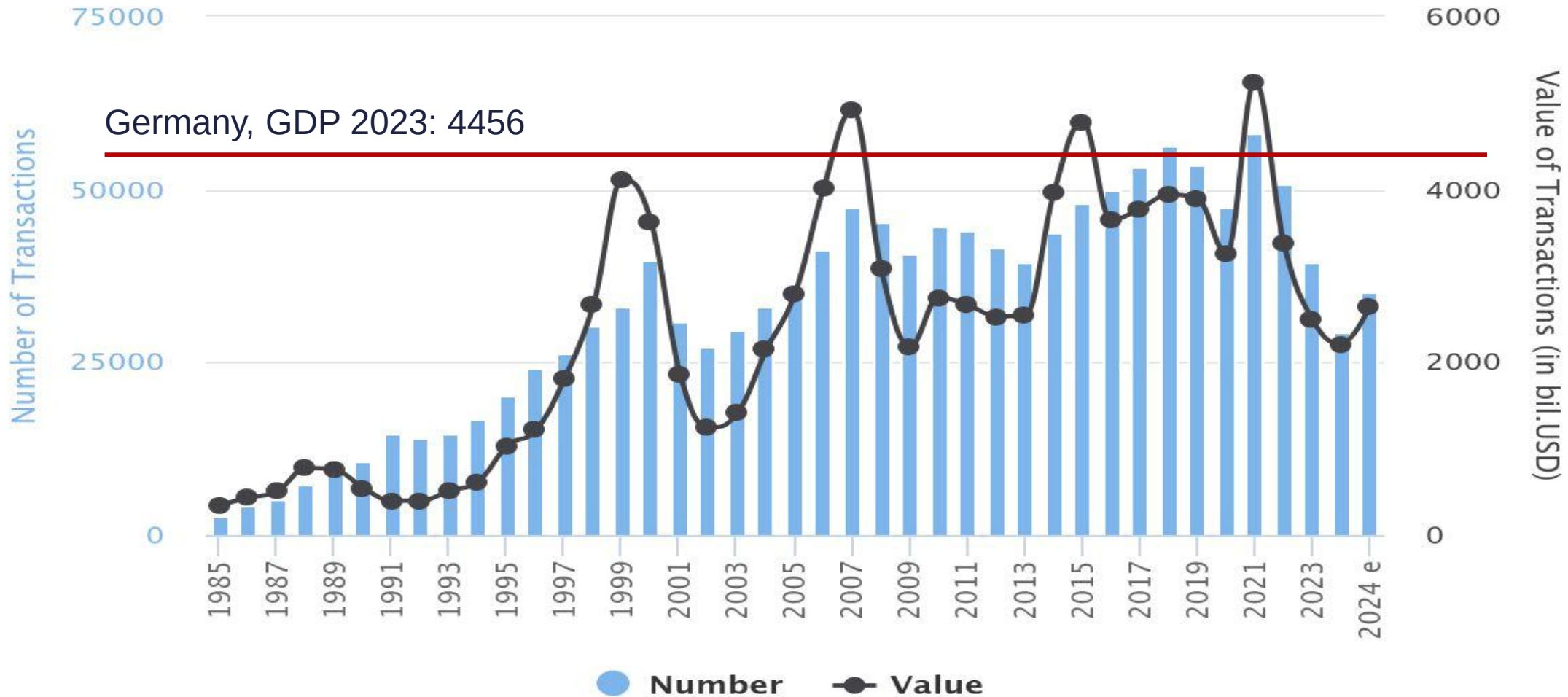
Tools:

New, positive NPV projects

Growth,

Sell-off of poor projects

Mergers & Acquisitions Worldwide



Sell-side motives

Management
issues
(inheritance)

Liquidity issues,
lack of capital
(bankruptcy,
growth)

Stress among
owners
(divorce)

Strategic
opportunities
(better investments
available)

exit
(venture capital)

Extensive growth

Efficiency of scale

Market force and
marketing considerations

Vertical integration

Diversification

Buy-side motives

Taxation

Access cash

EPS increase

Prestige

Complementary resources

Acquiring crown jewels

Undervaluation

Management considerations



Dilbert.com DilbertCartoonist@gmail.com



I-30-10 ©2010 Scott Adams, Inc./Dist. by UFS, Inc.



Do M&As create value?

For the sellers, absolutely.
The average takeover
premium is 30% in the US.

There is a huge deviation
across buyer effects –
usually they lose value.

Creating value by takeovers

Average Deal value added¹ – Target and acquirer, Percent

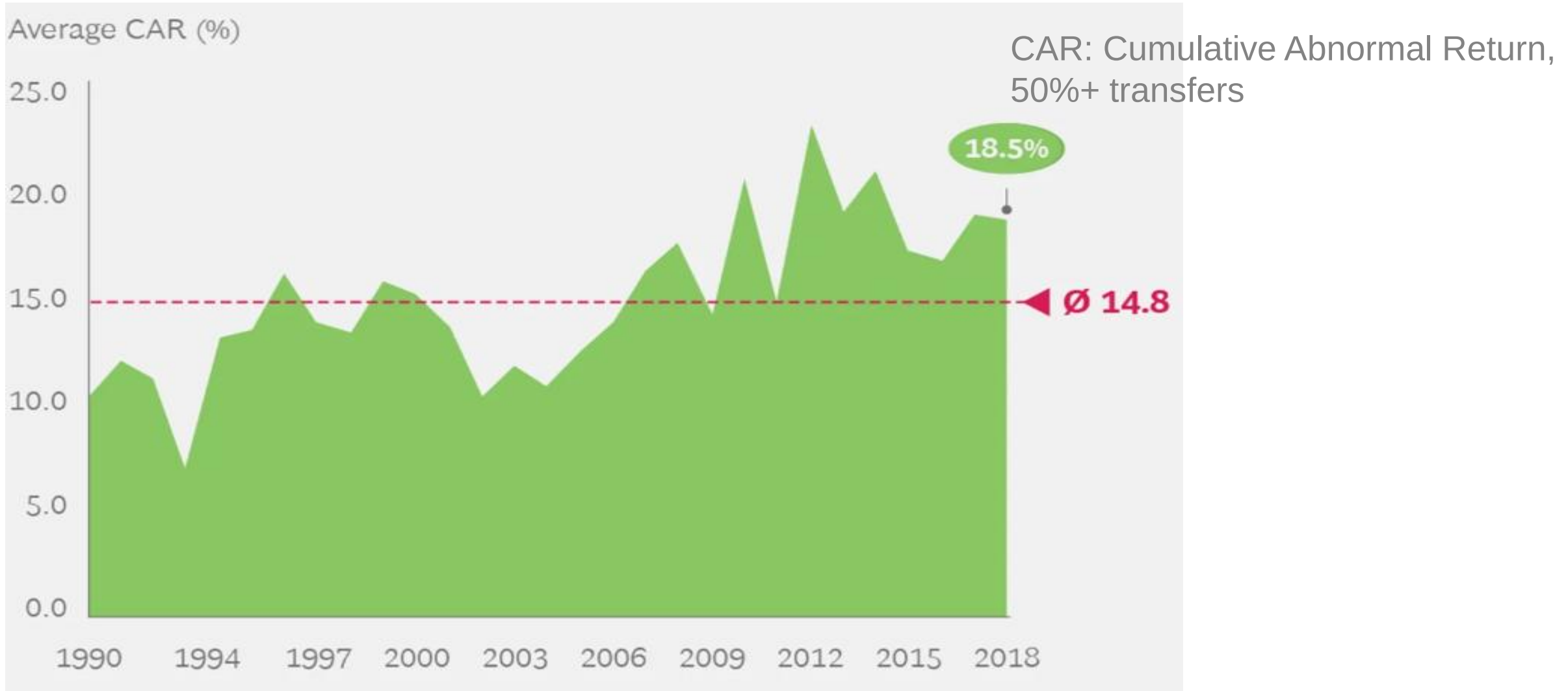
Acquiror DVA Target DVA Total DVA xx Number of deals



Source: McKinsey (2017): Overview of M&A, 2016

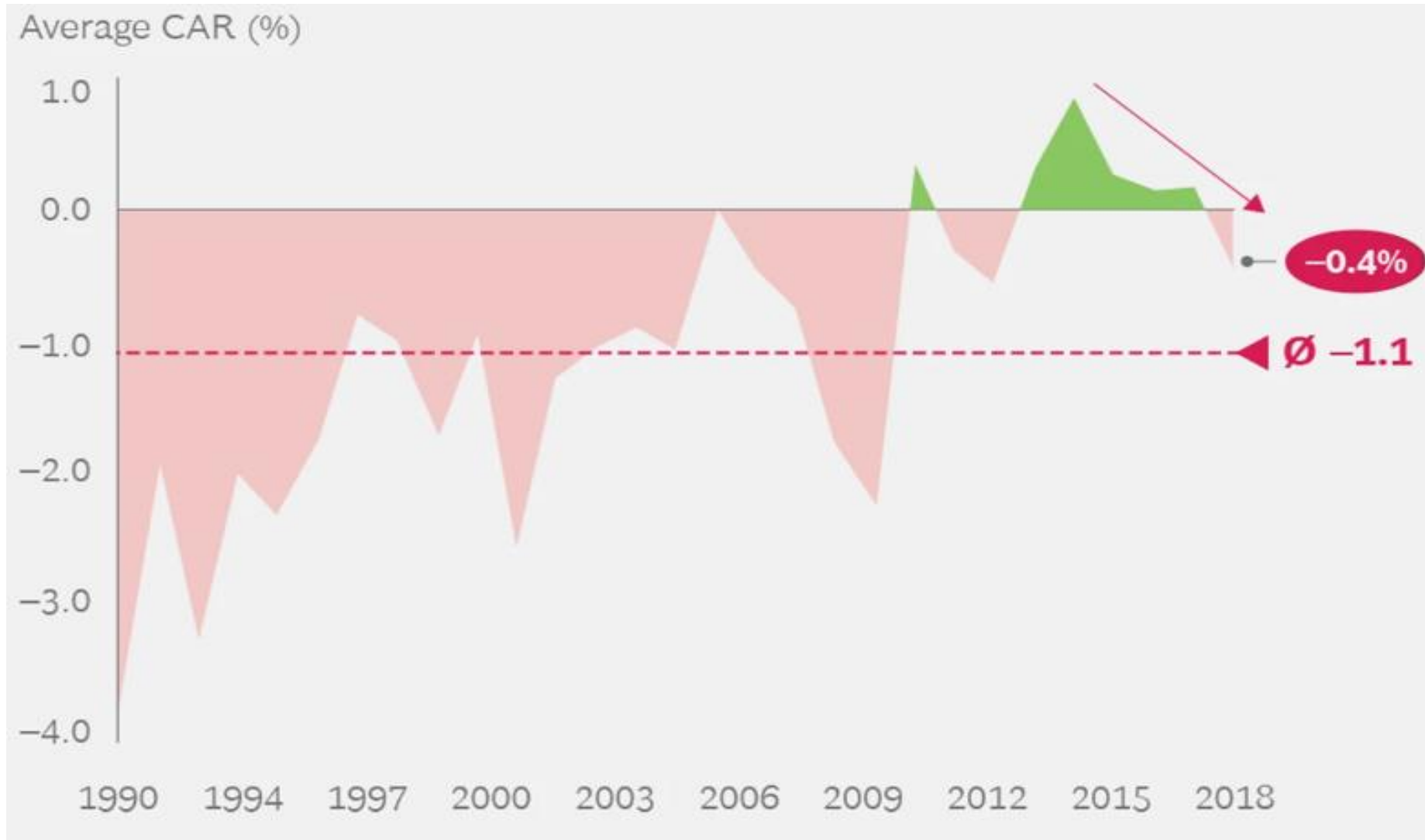
DVA: Deal Value Added, the change of joint market value of the target and the buyer, -2/+2 days around the announcement.

Creating value by takeovers (sellers)

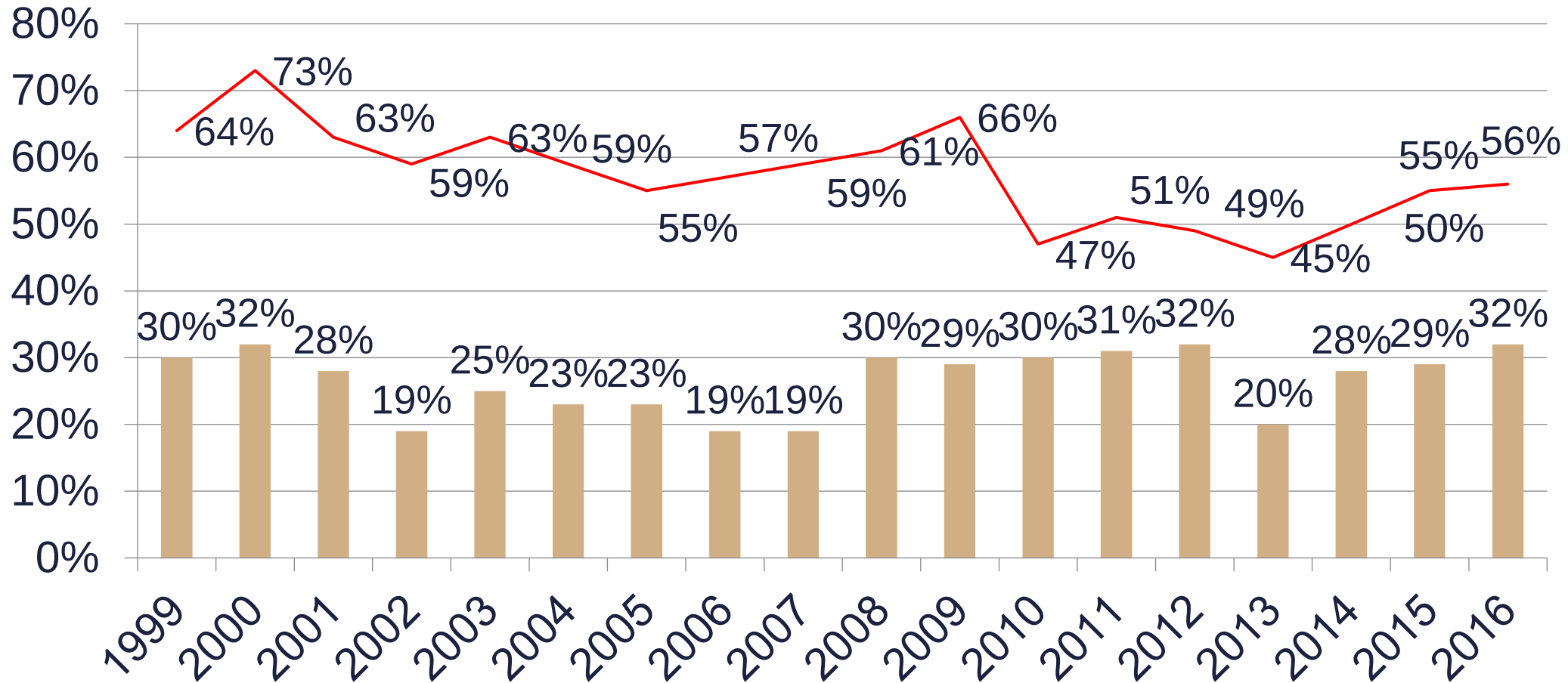


Source: BCG, 2019 Mergers & Acquisitions (M&A) Report

Creating value by takeovers (buyers)



Premiums on the global M&A market



Source: McKinsey, 2017, Premium: -7/0 days, median

POP: Percent of Overpayers, rate of buyers seeing their value falling due to the announcement, -2/+2 days around the announcement.

EV/EBITDA multiples

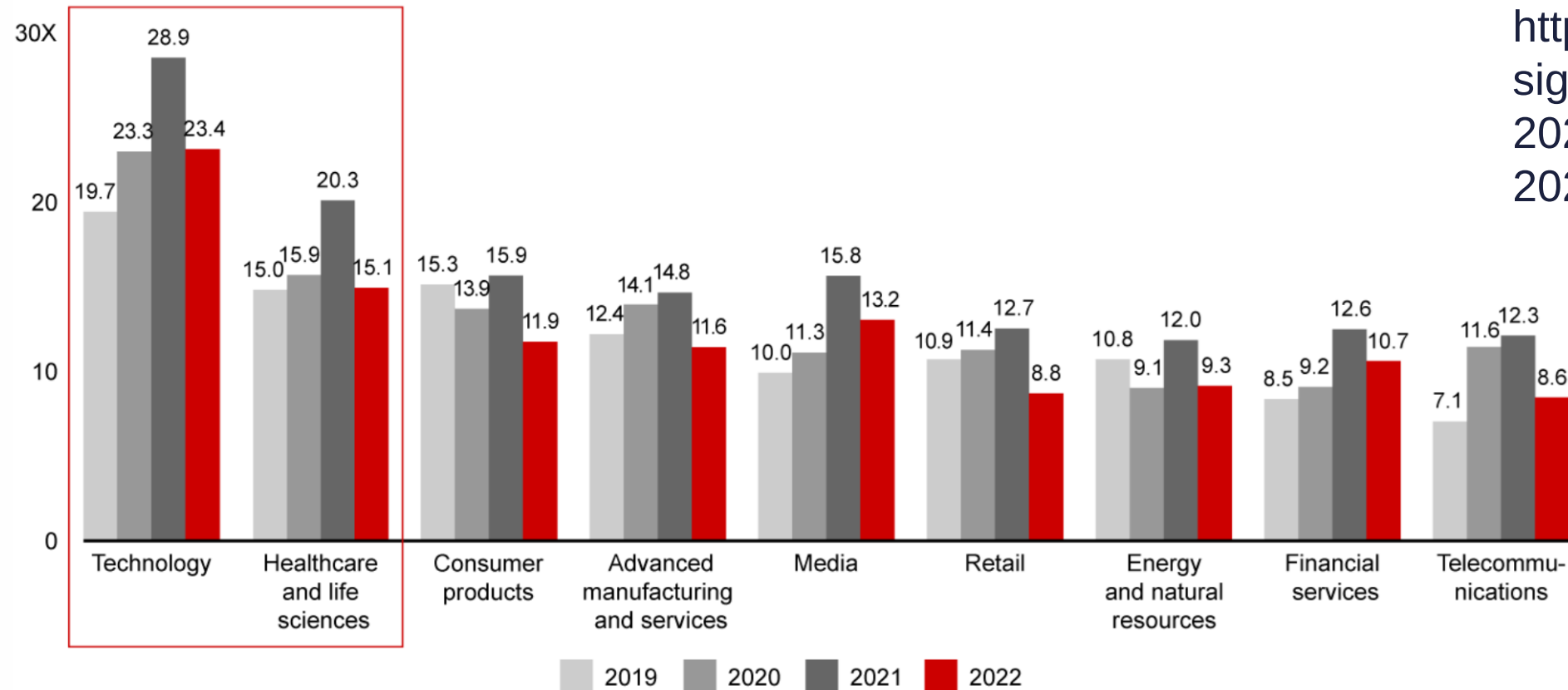
(global median)



Source: BCG, 2019 Mergers & Acquisitions (M&A) Report

EV/EBITDA multiples (global median)

Median enterprise value to EBITDA multiples per industry (strategic deals)

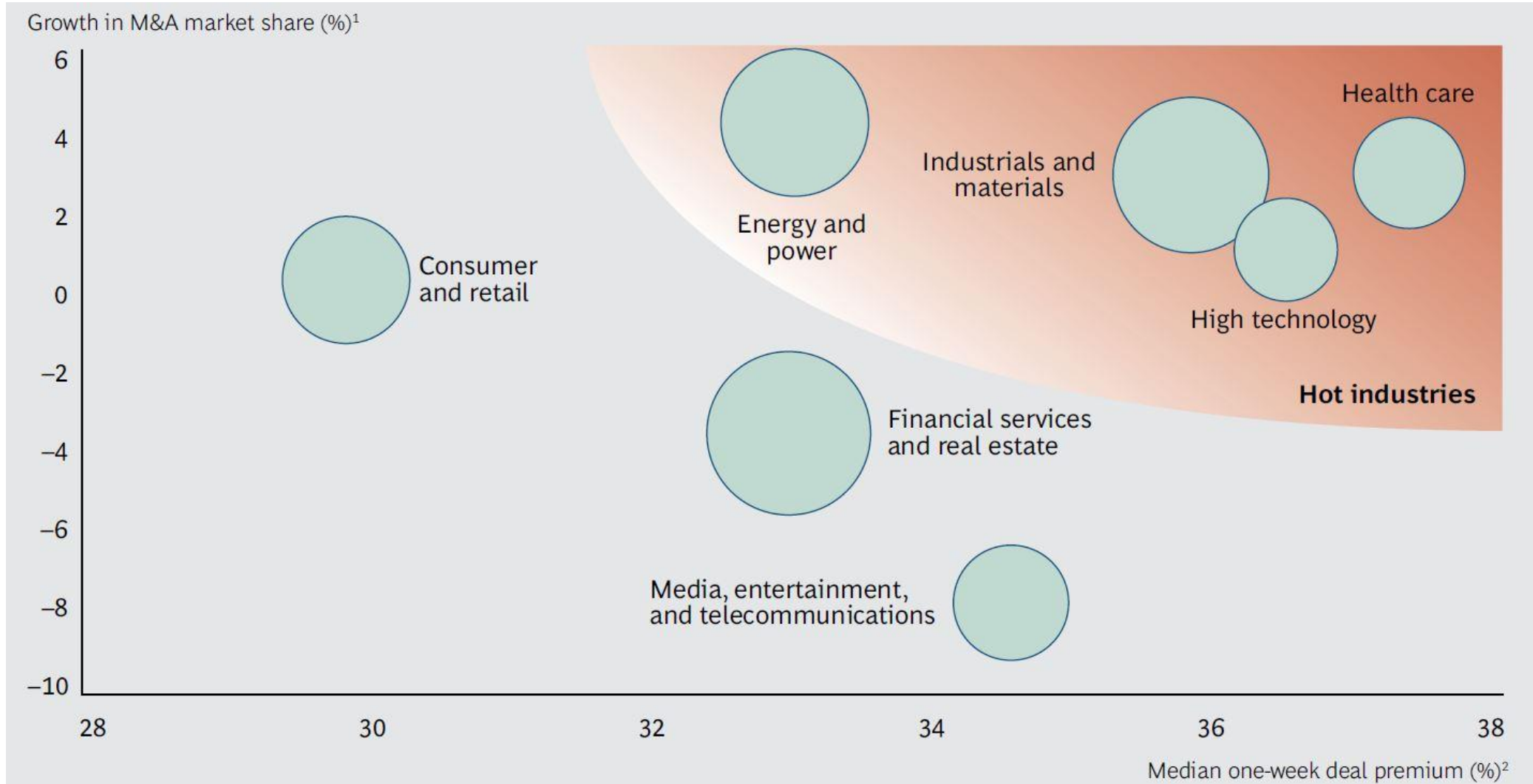


Source:

<https://www.bain.com/insights/looking-back-at-2022-m-and-a-report-2023/>

Notes: Median deal multiples for announced strategic deals in which valuation data was available; strategic deals include corporate M&A and PE portfolio add-ons
Source: Dealogic

M&A premium across industries



Source: BCG, 2014, 2011-2014 H1 transactions, median

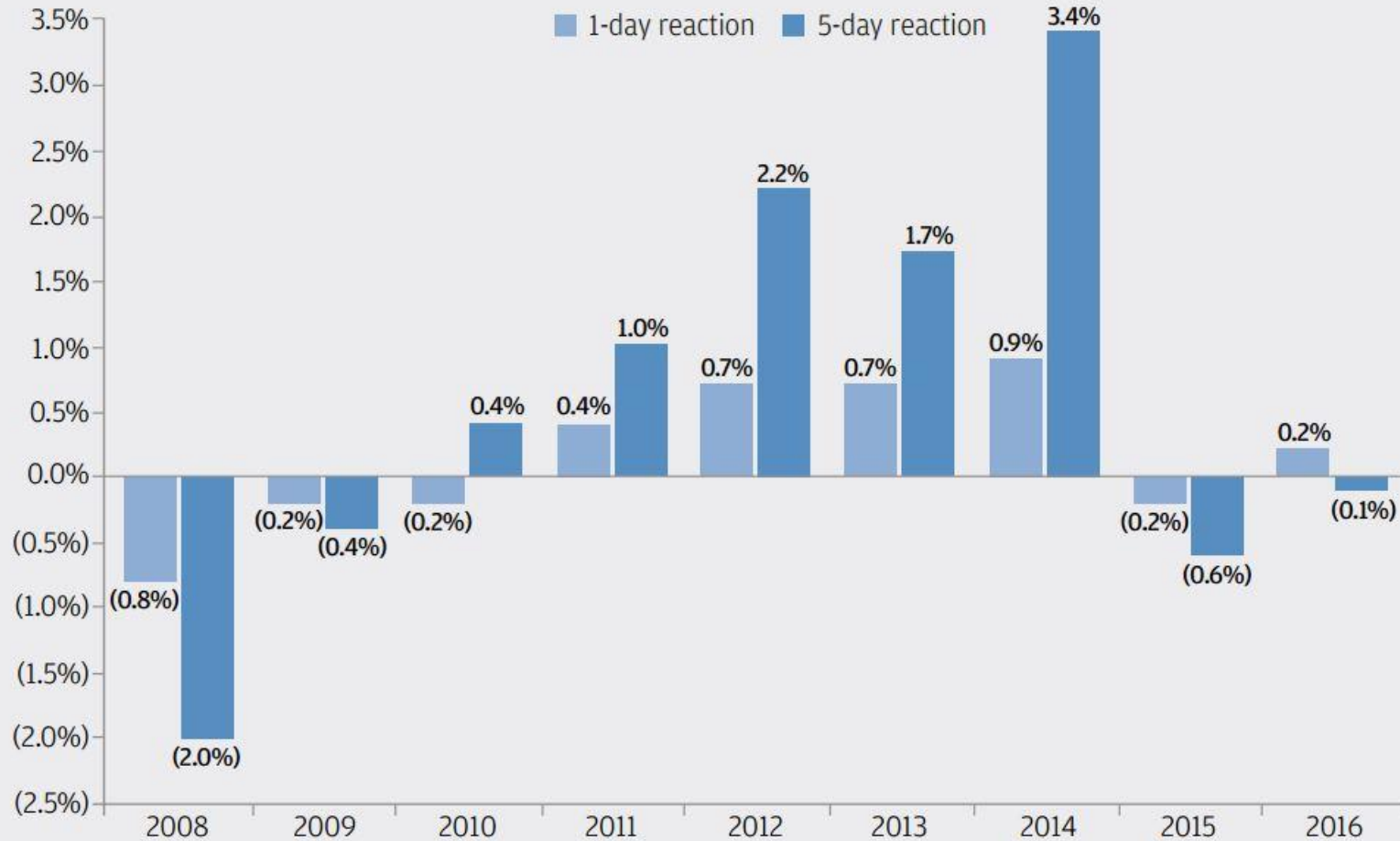
Creating value by M&As

Survey	Positive	Neutral	Negative
Based on financial ratios			
Mercer, 1995	25%	30%	45%
KPMG, 1999	17%	30%	53%
A.T. Kearney, 1999	27%	19%	54%
KPMG, 2001	30%	39%	31%
McKinsey, 2001	12%	46%	42%
Based on management interviews			
MAPI, 1999	61%	27%	12%

Source: Pautler, 2003

Shares price reaction (buyers)

Source: JP Morgan,
2017, median, US



Sources: FactSet and company filings as of December 31, 2016

Note: Includes acquisitions by U.S. buyers with minimum deal value of \$500mm, % owned < 20%, % acquired > 80% and target > 5% of the size of the acquirer

^a Excess return over S&P 500 returns times acquirer's beta from unaffected date prior to announcement

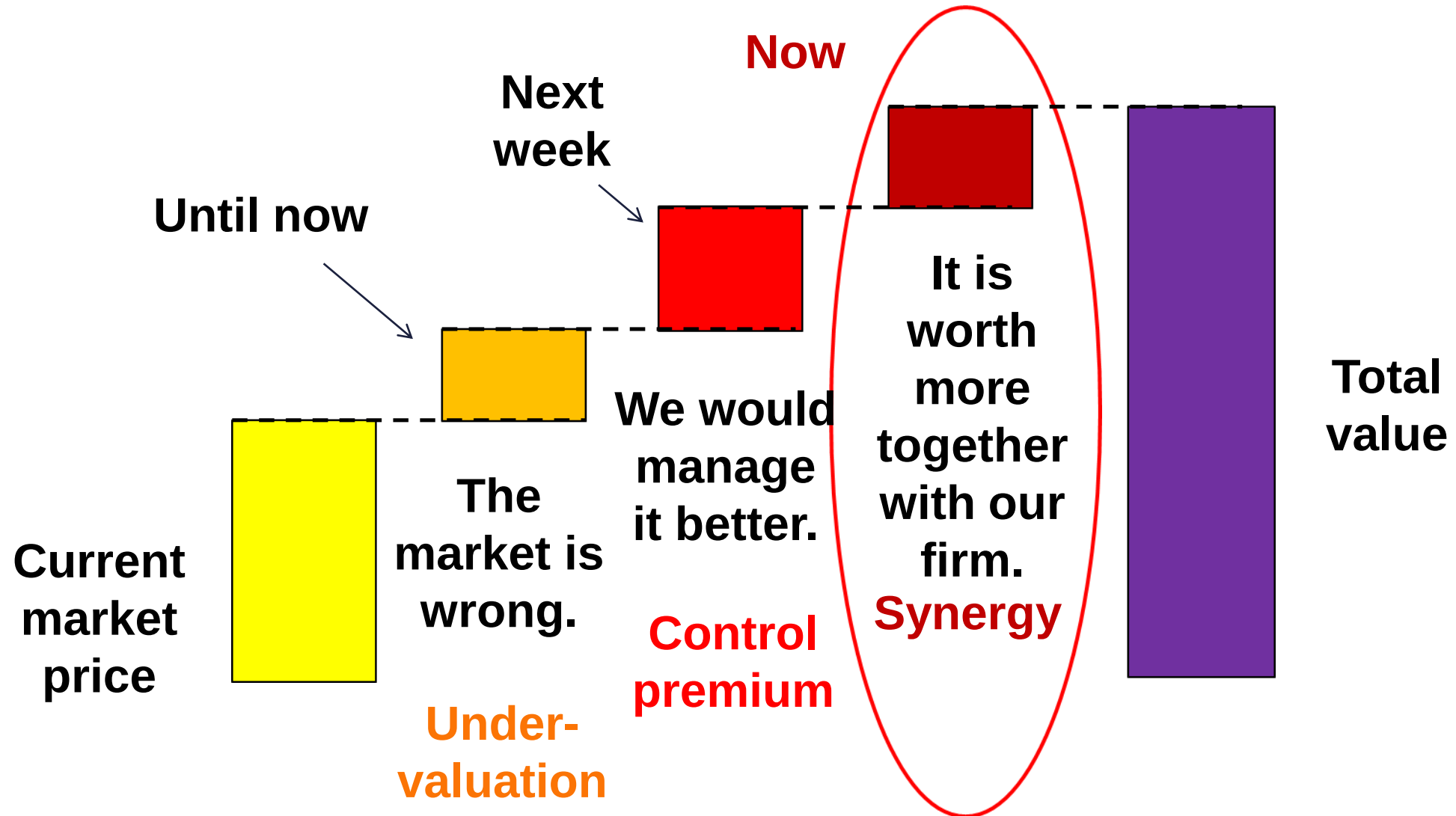
Why paying more than the current price?

Undervaluation „premium”: The fair firm value is 3 today, but the market price is only 2. We pay a maximum of 3; no need to own another firm.

Control premium: Should we control the firm, the value will climb from 2 to 3. No need for owning another firm.

Synergy: Two firms together add up for more: $2+2=5$, thus we pay a maximum of 3 for the firm.

The structure of the total value



What is synergy?

The effect due to which parts
together are
more valuable than separately.

Car – ignition key

Cell phone – SIM card

Computer – software



Operational

Synergies

Financial

Synergies

Strategic

**Economies
of scale**

**Free
cash**

**Increased
debt
capacity**

New
projects
with higher
profitability

Extra
return over
longer
period

More
new
projects

Cost
savings

**Tax
reductions**

Return
and $g \uparrow$

Higher g

Longer
growth
period

NOPLAT and
operational
margin $\uparrow$

Total NPV of
project available $\uparrow$

Cost of
capital $\downarrow$

NOPLAT $\uparrow$

What form do those show?

How do those affect value drivers?

When do those start having an impact?

Modelling operational synergies (higher FCFF):

Cost efficiency – naturally limited

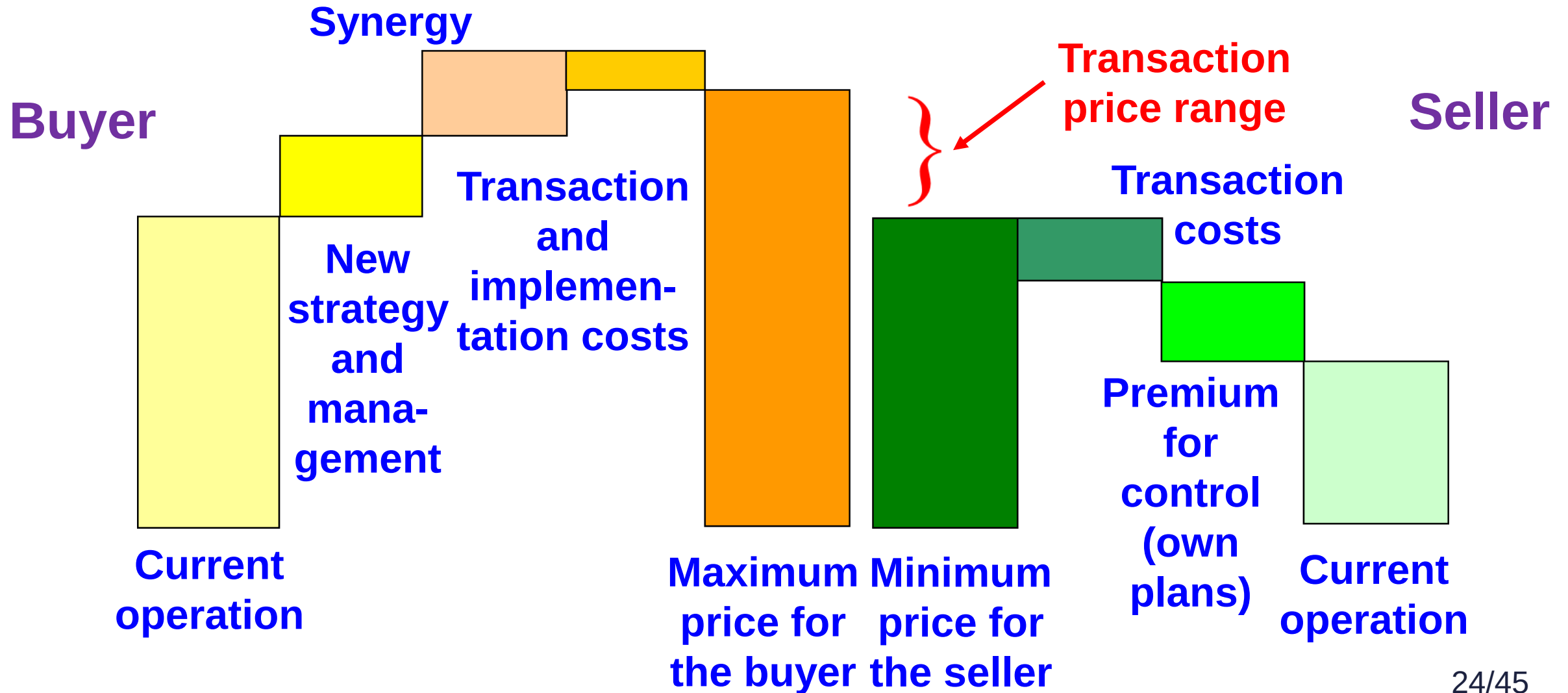
Sales growth synergies – only limited by common sense (Country and market growth? Maintainable?)

EPS growth – if a firm with a $P/E = 30$ buy another with $P/E = 20$, EPS would grow (And what then? – this is not a value focused approach)

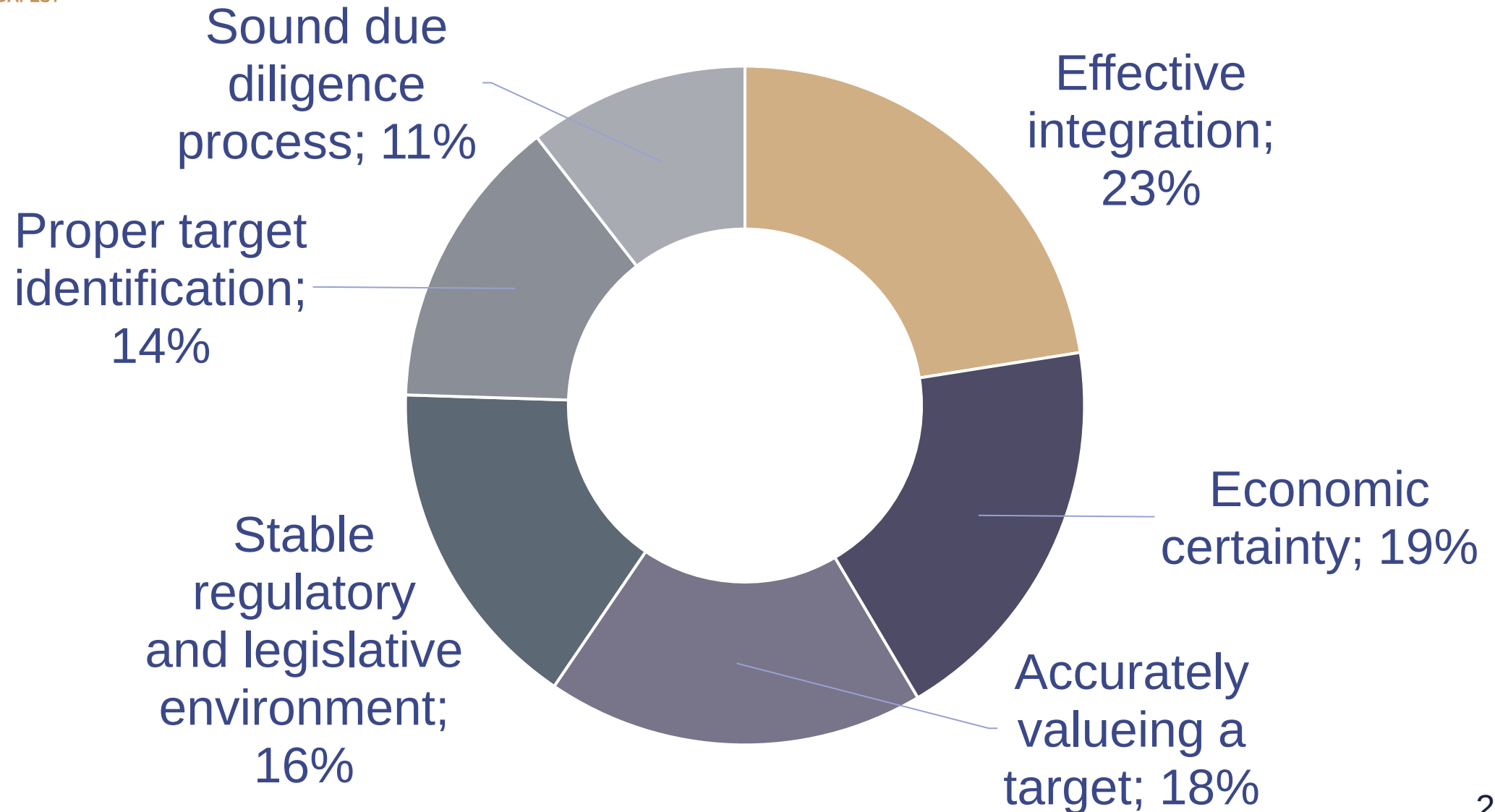
Faster growth – it is of no value alone. We need to review:

- $WACC \leftrightarrow ROIC$, $RONIC$?
- Premium $\leftrightarrow$ Synergy?

Setting the price



Success factors of M&As



Why would mergers fail?

1. Overpayment (extreme optimism)

Overoptimism on market opportunities

Overestimating synergies

Incomplete due diligence, hidden or unhandled issues

Overbidding other offers, management ego („the winners’ curse”)

2. Poor integration management

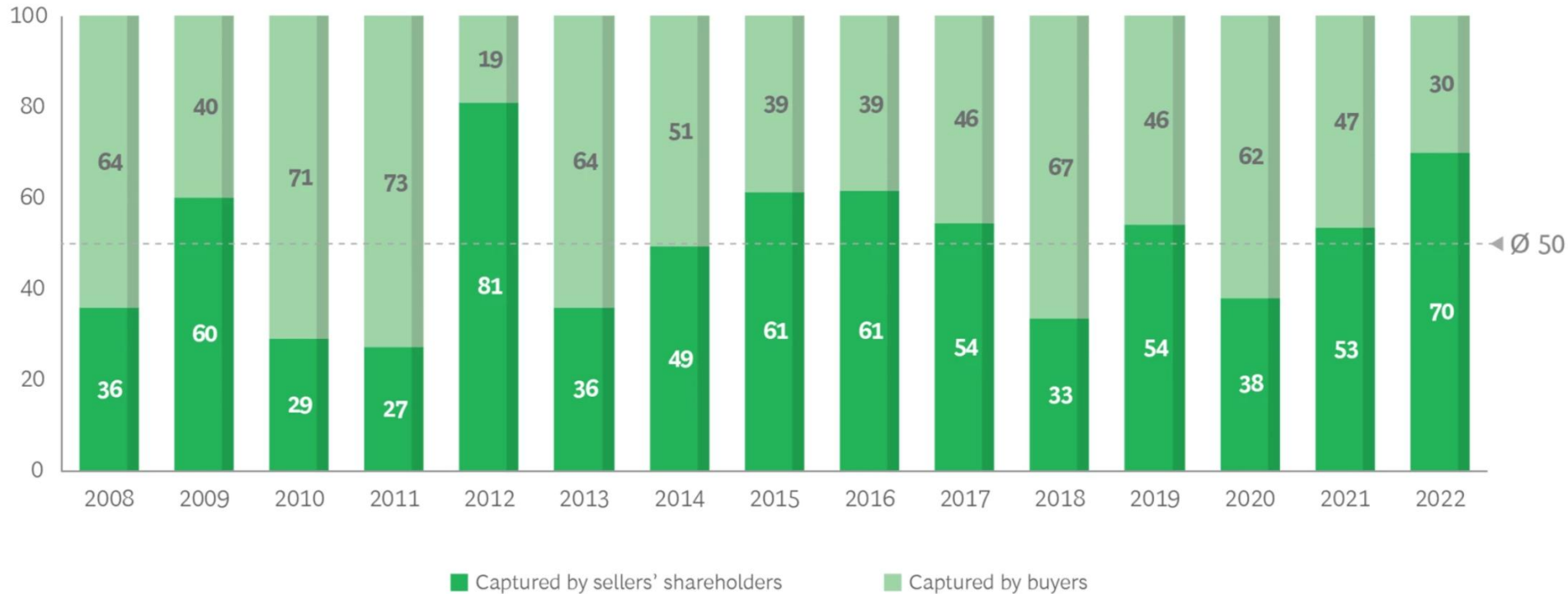
Lack of setting a new business model

Uncertainties and conflicts remain unsolved

Late reactions to market pressures

Premium vs Synergy

Median value of synergies captured by sellers' shareholders and buyers (%)¹



Sources: Refinitiv; company press releases; BCG analysis.

Note: Analysis based on 1,385 public-to-public M&A transactions from 2008 through 2022 with a majority stake acquired by nonfinancial buyers.

¹Value captured by sellers' shareholders is based on cumulative abnormal returns during the period from three days before to three days after announcement of the deal.



www.dilbert.com scottadams@aol.com



12-18-03 © 2003 Scott Adams, Inc. Dist. by UFS, Inc.

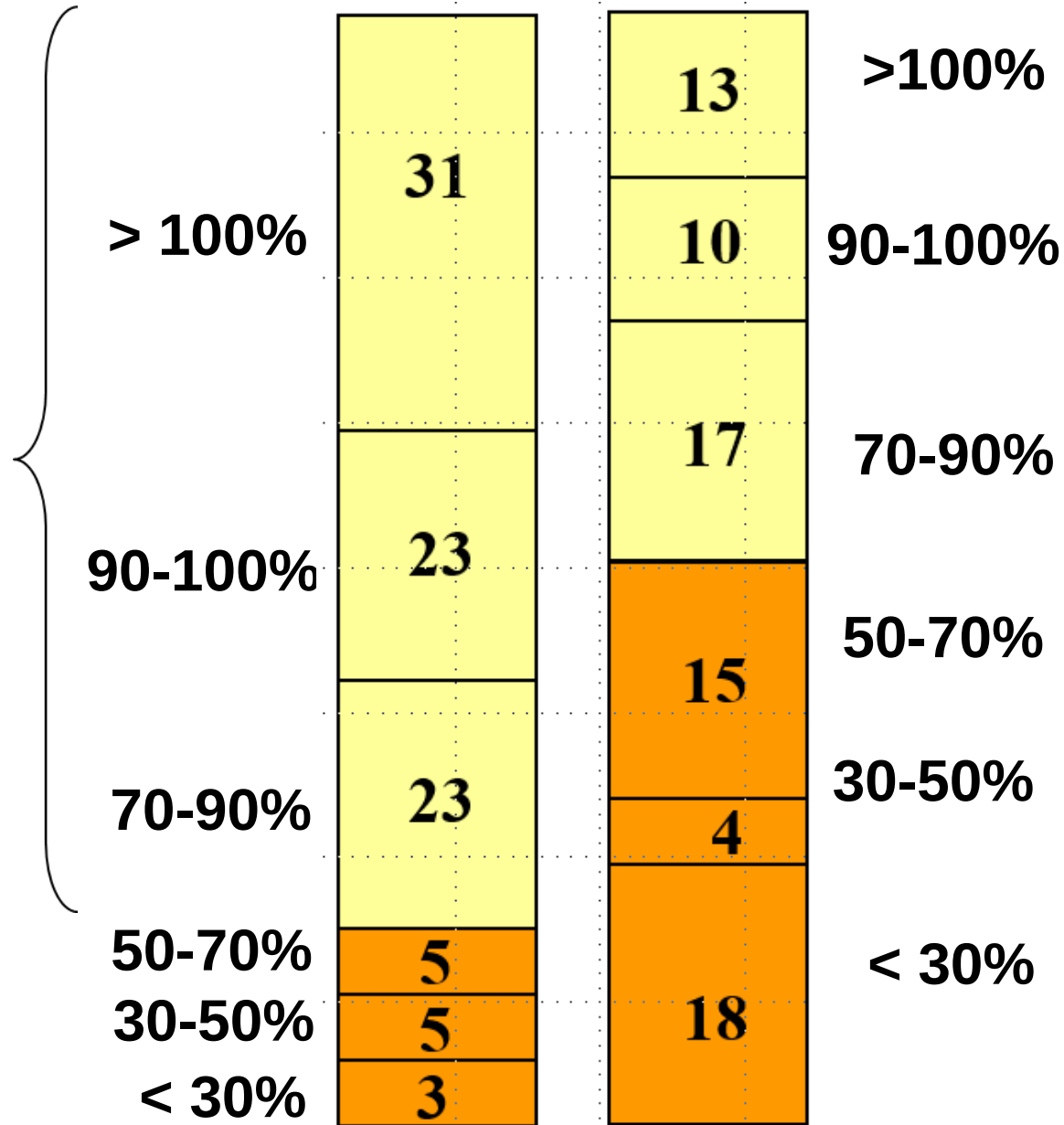


In **85 %** of the transactions, at least 70% of cost synergies were achieved.

N=90

Cost synergy

Source: McKinsey

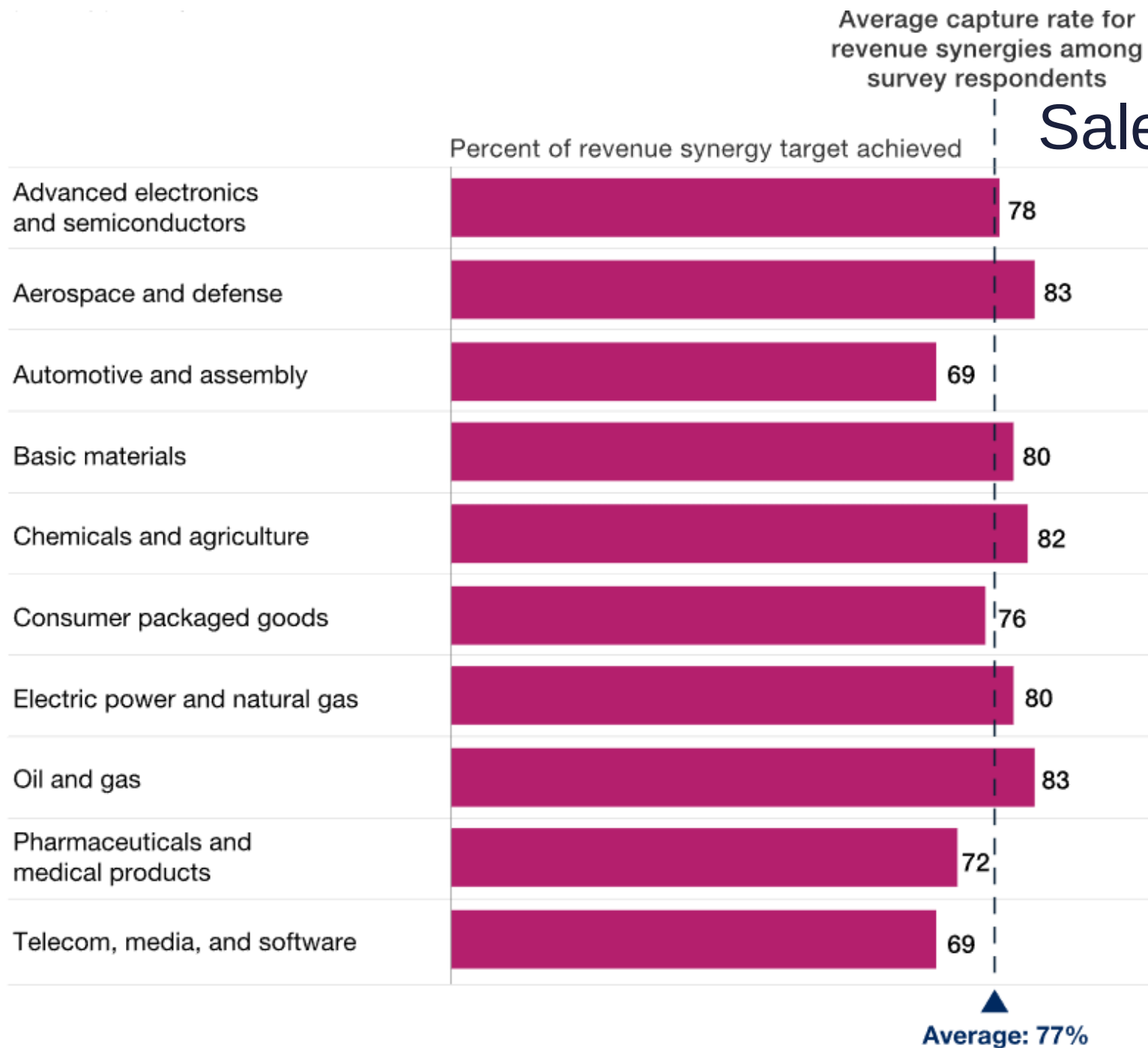


Plans vs Reality

In **52%** of the transactions at least 70% of the sales synergies were realised.

N=77

Sales synergy



Sales synergies realised

Source: McKinsey, 2018

The likelihood of realising the different types of synergies is not the same.

The joint sales will not start growing immediately as

- 1) the transaction process jeopardises the customer connections.
- 2) competitors aim immediately at our best staff,
- 3) customers buying from both firms hit take some of their purchase to a competitor,
- 4) the remaining customers ask for a reduction as thanks to their loyalty

Costs of realisation

lay-off expenses, restructuring, repositioning, change of names, IT integration

These can add-up to more than one year of synergy

Time need of realisation

Different customer management systems, trainings, IT integration

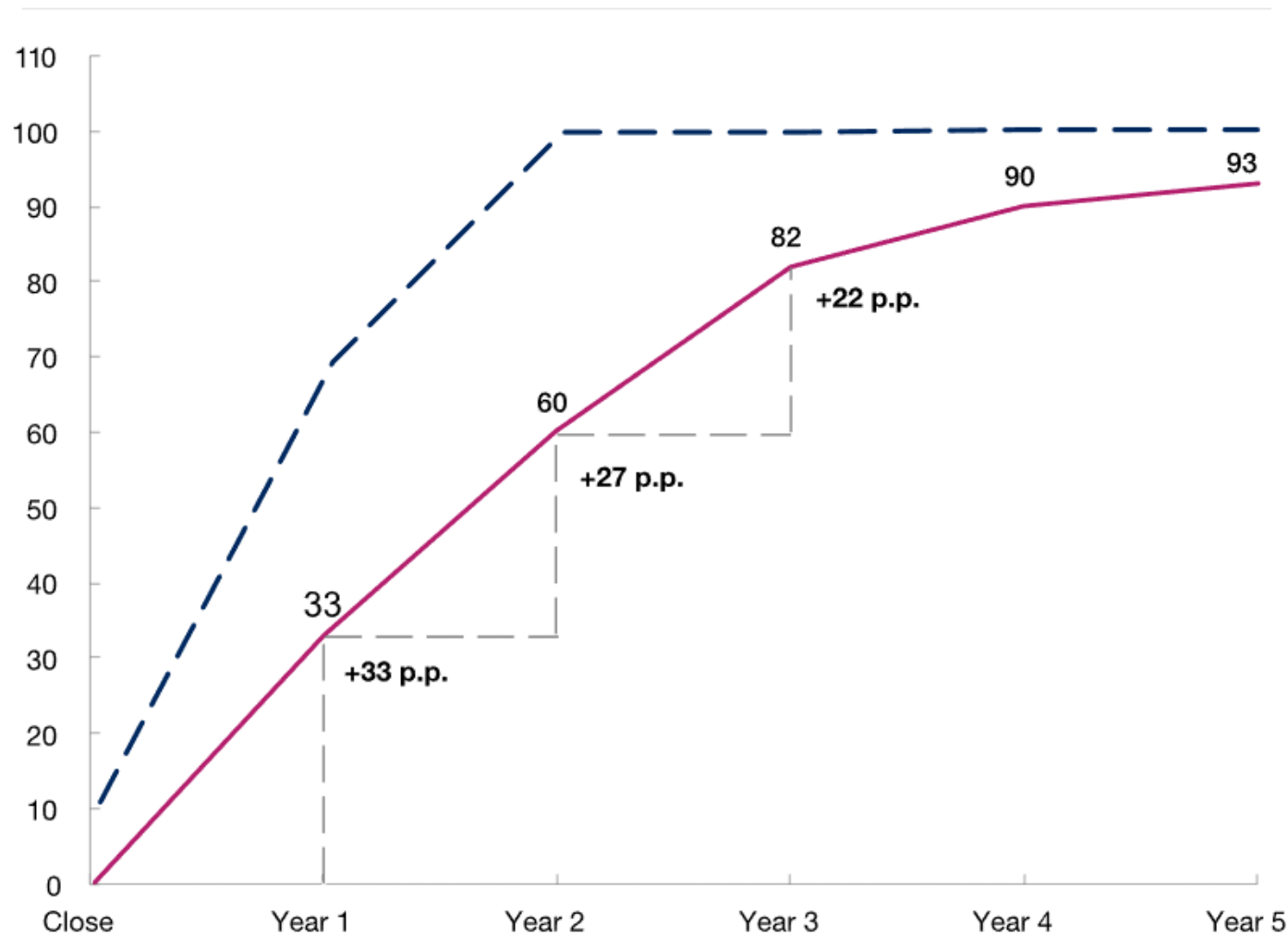
Synergies that could not be realised during the first complete business year after the transaction are usually completely lost.

The time need of realising the synergy

Percent of total synergy target captured by the end of each year after deal close

— Cost synergies (illustrative)

— Average revenue synergies



How to pay for a firm?

Cash



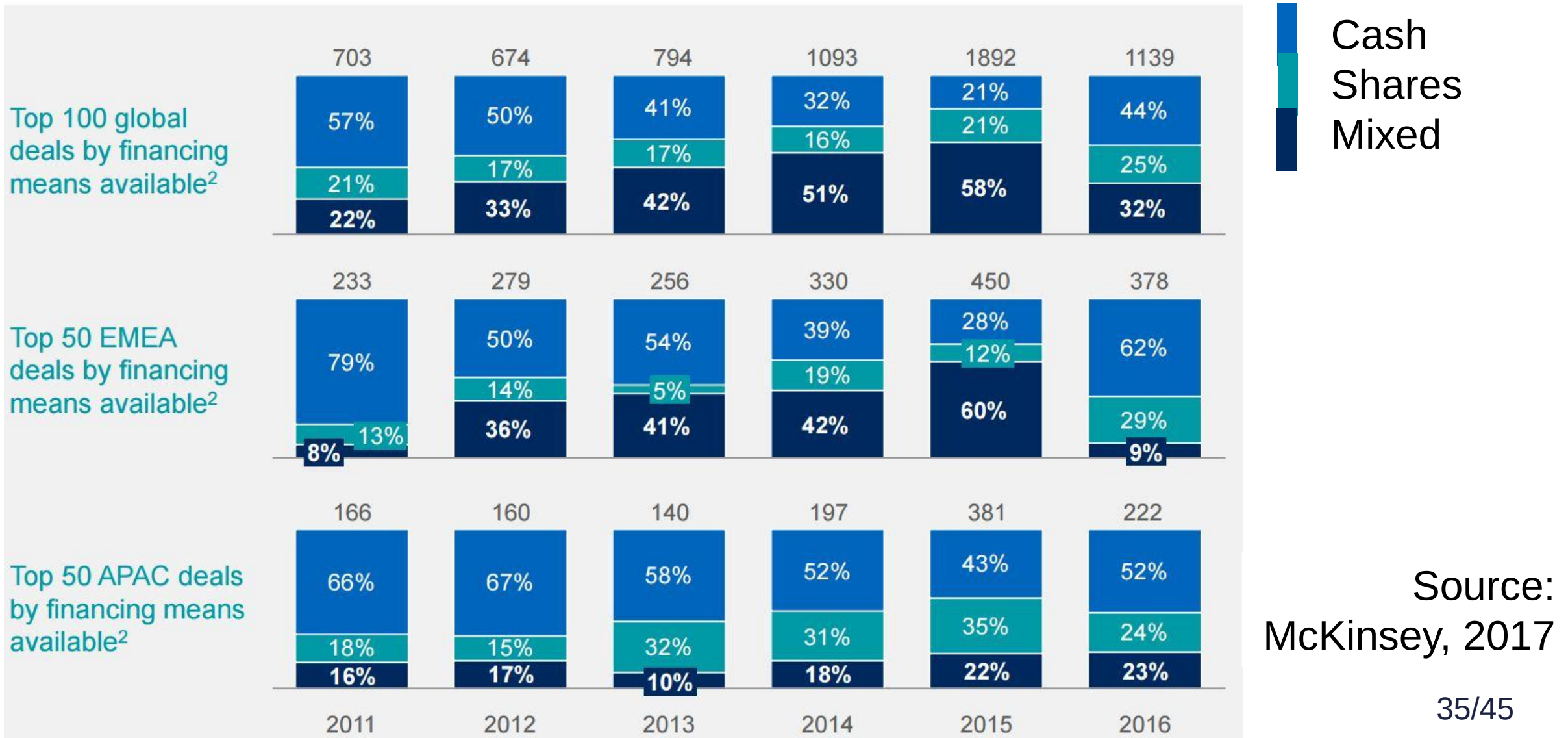
- Liquidity issues
- All risk remains with the buyer
- All profit stays with the seller

Shares

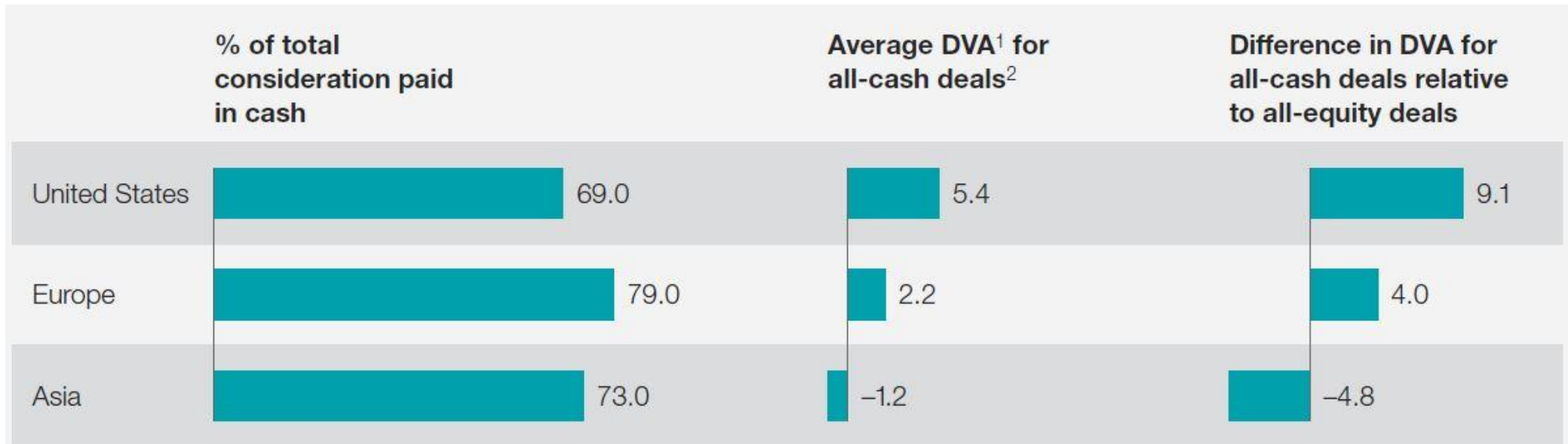


- The seller can not exit
- Shared risks
- No liquidity shock

Payment methods worldwide



Market view of payment methods



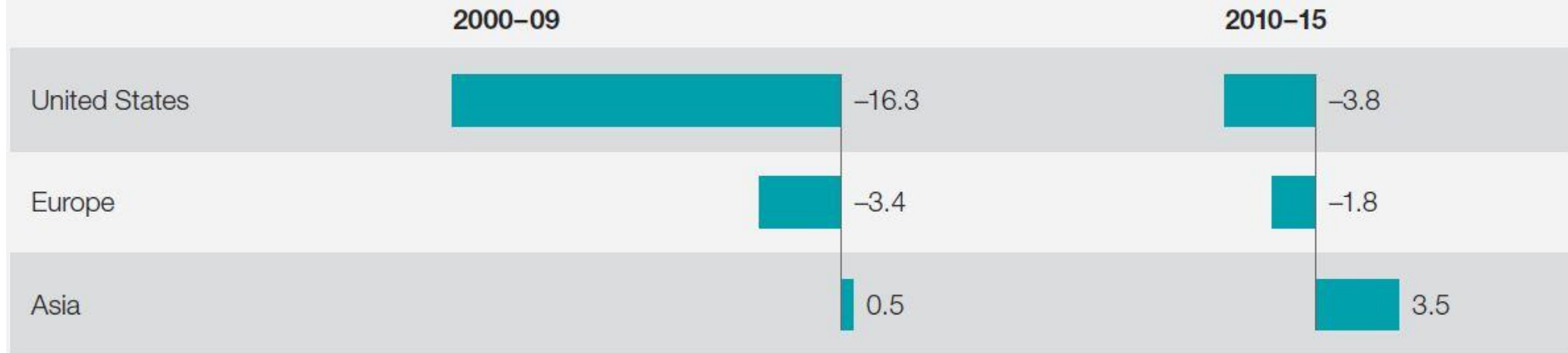
¹ Deal value added.

² For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

Market view of shares-only payment

Average DVA¹ index for all-stock deals,² %



¹ Deal value added.

² For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

1. The fusion of **similar-sized firms** is less likely to be successful.
2. Synergies from **immediate cost reductions** (e.g. downsizing) are easier to capture than those linked to future growth, new products, or joint R&D.
3. **Hostile takeovers** promise better management and results than friendly transactions.

Lessons from the history 2

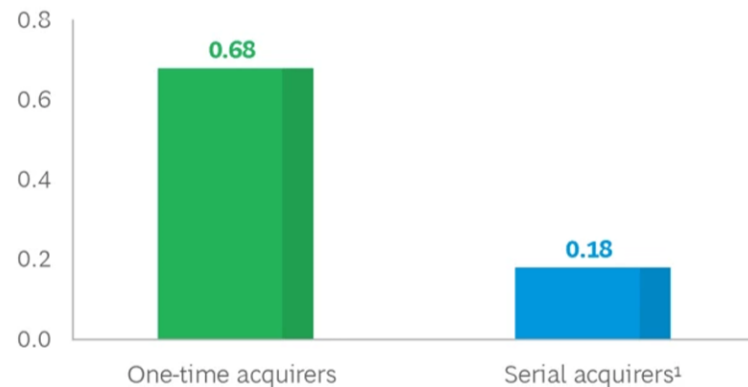
3. Acquisitions focused on **smaller private firms** are more likely to be successful. The fiercer the competition is, the more likely the loss in value.
4. **Firms carefully considering** synergies and M&As are more likely to be successful.



5. **Companies with outstanding efficiency** are more likely to create value by acquiring firms.
6. **Firms that gained experience in M&A** are more likely to be successful in the long run.

Less-experienced dealmakers create a strong initial pop...

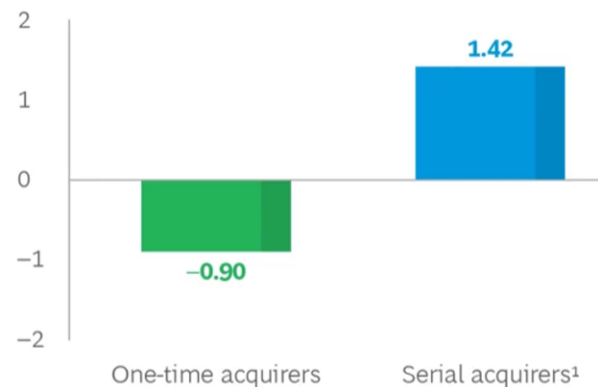
Cumulative abnormal return (%)



Acquirer experience level

...but fail to create long-term value

Two-year relative TSR (%)



Acquirer experience level

Source: BCG,
2023 Mergers &
Acquisitions
(M&A) Report

**Thank you
for your attention.
Any questions?**

peter.juhasz@uni-corvinus.hu



ABC Co. purchases BCD Co. They pass on 100 unit of synergy,
one third of the price is paid cash.

How will the price of ABC shares change due to the transaction, if
there are 2000 pieces issued?

ABC current value: $V=1000$ $E=800$ $D=200$

BCD current value: $V=300$ $E=200$ $D=100$

Synergy=300

Joint value (before payment): $V=1600$ $E=1300$ $D=300$

BCD shares purchase price: $200+100=300$

Cash: $300 \cdot \frac{1}{3} = 100$

Shares: $300 \cdot \frac{2}{3} = 200$

Together (after the payment): $V=1500$ $E=1200$ $D=300$

ABC current quote: $800/2000=0.4$

Synergy to current owners: $300-100=200$

Required values of ABC shares after the transaction:

$$800+200=1000$$

Share price: $1000/2000=0.5$

To be paid for BCD: 200

Number of shares to issue: $200/0.5=400$

Number of shares after the transaction: $2000 + 400 = 2400$

Price quote: $1200 / 2400 = 0.5$

Attention! If during the transaction the amount of debt changes of extraordinary dividend is paid, you also need to consider those effects before calculating the share prices and number of shares to issue!